

Envisioning Identity: The Social Production of Human-Centric Computer Vision Systems

Morgan Klaus Scheuerman
Information Science, University of Colorado Boulder
morgan.scheuerman@colorado.edu

ABSTRACT

Computer vision technologies have been increasingly scrutinized in recent years for their propensity to cause harm. Human-centric computer vision, systems designed to interpret visual data about humans for a variety of tasks, are perceived as particularly high risk. Broadly, the harms of human-centric computer vision focus on demographic biases (favoring one group over another) and categorical injustices (through erasure, stereotyping, or problematic labels). Prior work has focused on both uncovering these harms and mitigating them, through, for example, better dataset collection practices and guidelines for more contextual data labeling. This research has largely focused on understanding discrete computer vision artifacts, such as datasets or model outputs, and their implications for specific identity groups or privacy. There is opportunity to further understand how human identity is embedded into human-centric computer vision not only across these artifacts, but also across the network of human workers who shape computer vision systems. My dissertation focuses on understanding how human identity is conceptualized across three different “layers” of computer vision: (1) at the artifact layer, where the classification ontology is deployed, in the form of datasets and model inputs and outputs; (2) at the development layer, where social decisions are made about how to implement models and annotations by traditional tech workers; and (3) at the annotation layer, where technical specs are applied, often by human data workers. I will highlight where identity across these layers do not align and, as a result, are not necessarily predicting identity accurately or even as intended by practitioners.

CCS CONCEPTS

• Human-centered computing; • Empirical studies in collaborative and social computing; Social and professional topics; • User characteristics;

KEYWORDS

Computer vision, identity, work practice, data work, technology companies

ACM Reference Format:

Morgan Klaus Scheuerman. 2022. Envisioning Identity: The Social Production of Human-Centric Computer Vision Systems. In *Companion Computer Supported Cooperative Work and Social Computing (CSCW’22 Companion)*,

November 08–22, 2022, Virtual Event, Taiwan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3500868.3561396>

1 INTRODUCTION

Identity, the qualities that define each individual, is salient to every aspect of the human experience. Identity can be theorized in a myriad of ways and is often viewed as a culmination of both the mind and the body. The worldview that one develops as a result of this intricate web of identities is often called *subjective positionality*—how the position a subject occupies in the world shapes their experiences and perspectives, and thus decision-making processes and agency that subject may have access to in specific contexts [1., 9., 18.].

The use of human identity characteristics in both the inputs (the training data) and the outputs (the inferences) of computer vision have been increasingly scrutinized by computing researchers (e.g., [6., 7., 14.]). Race and gender have become two of the largest concerns regarding bias in machine learning fairness literature—particularly, how systems are biased against certain races and genders [3., 8., 12.] and how to mitigate those biases [4., 5., 20.]. Much of this current research on ethics in computer vision has focused on technical artifacts—the datasets (e.g., [13.]), the documentation (e.g., [11.]), and the models (e.g., [2.]).

There has been less research on the actual work processes and positionalities of those involved in shaping computer vision, resulting in obscurity about the role of human subjectivity in defining identity in computer vision. In terms of work practices around computer vision, in the case of either traditional tech workers or data workers¹, Miceli et al. highlight the power that traditional tech workers enact over the data annotation process [10., 11.]. Not only do traditional tech workers define the taxonomies underlying data labeling in ways that reflect their own specific worldviews regardless of the cultural site of data annotation [10.], annotators regularly view traditional tech workers as having more expertise and knowledge than themselves, and therefore do not question labeling guidelines [11.]. Such prior work on data workers has highlighted a need to better understand the role of workers, including both traditional tech workers and data workers, in shaping the subjective outcomes of machine learning, including computer vision.

It is critical to understand the *social* production of computer vision artifacts by examining how the positionalities of tech workers in the many roles across the computer vision development pipeline embed their values and perspectives into them. Thus, in this dissertation, I plan to more deeply explore how identity is embedded into computer vision and how the humans involved in the processes

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW’22 Companion, November 08–22, 2022, Virtual Event, Taiwan

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9190-0/22/11.

<https://doi.org/10.1145/3500868.3561396>

¹Often “gig workers” who work on short term contracts, or “gigs” [19.]

of computer vision development shape these identity-based outcomes in human-centric computer vision systems. I propose that the “low-level” decisions of individual workers is impactful to the end product of computer vision artifacts, and should be viewed as a site of intervention beyond solely high-level policies or guidelines. To demonstrate how identity is conceptualized in computer vision, I specifically aim to answer the following research questions:

- How is identity embedded into computer vision and what does that communicate about the values of the field?
- How do industry professionals working on computer vision pipelines make decisions about incorporating human identity into computer vision systems?
- How does individuals’ positionality impact the way they do their work and how does that influence the outcomes of computer vision models?
- What are the relationships that shape and constrain workers when developing computer vision models?

My work will assess computer vision infrastructures, not just as sociotechnical systems, but as inherited and interconnected layers with the goal of understanding how they reference, leverage, and constrain one another. I will connect the dots of positionality across the computer vision pipeline, from traditional tech worker to data worker to the final product, both dataset and model.

2 COMPLETED WORK

The focus of my completed work is to showcase how datasets and models look like in practice currently. I will present the works (1) *How We’ve Taught Algorithms to See Identity: Constructing Race and Gender in Image Databases for Facial Analysis* [17.]; (2) *How Computers See Gender: An Evaluation of Gender Classification in Commercial Facial Analysis Services* [16.]; and (3) *Do Datasets Have Politics? Disciplinary Values in Computer Vision Dataset Development* [15.]. The first completed study examines how identity labels—specifically labels for “visible identities” like gender and race—are produced for human-centric computer vision datasets. The second completed study interrogates how computer vision models produce specific worldviews about identities, in this case gender. The third completed study examines computer vision datasets at a higher level, focusing on what computer vision artifacts like datasets communicate about the values of the field. How identity and positionality is represented in artifacts produced for computer vision communicates the values of the end products produced by computer vision practitioners. Synthesizing and discussing this work allows me to set up the proposed work as necessary to understand human-centric computer vision practices in *reality*, not as interpreted through existing artifacts.

3 PROPOSED WORK

Having completed the three studies above, I possess unique insight of the underlying politics embedded into human-centric computer vision artifacts, how the artifacts themselves shape identity. However, the completed work does not explicate the decisions that those developing computer vision make when embedding identity into artifacts. To extend on this prior work, I have designed three studies

to focus on how identity shapes artifacts; how practitioners working on computer vision navigate interwoven values in the context of labor roles.

4 STUDY ONE: UNDERSTANDING HOW TRADITIONAL TECH WORK SHAPES IDENTITY ATTRIBUTES IN COMPUTER VISION

The role that traditional tech workers—in big tech and in startup settings—play in the development of computer vision technologies is opaque. The goal of the first study is to understand how identity is conceptualized during the development and maintenance of computer vision products that intersect with human identity—whether explicitly (e.g., classifying aspects of human faces, like gender or emotion) or implicitly (e.g., gesture recognition systems which must recognize diverse hands). To accomplish this, I will conduct semi-structured interviews with between 15-20 tech workers at both smaller companies and large companies. After completing interviews, I will transcribe and then complete a thematic analysis and detailed memoing on the text transcripts (Salmons, 2018). Analysis will focus on uncovering emergent themes about how traditional technology laborers think about, reason through, and shape the implementation of identity in computer vision products. Findings will contribute a preliminary understanding to the role that human laborers play in actively shaping the computer vision model pipeline, from data to output. At a higher level, themes emergent in this phase of the proposal will be analyzed alongside findings from prior work and the following proposed work to develop a theory on how computer vision—and AI more broadly—is developed across a complex network of human and technical actors.

5 STUDY TWO: UNDERSTANDING HOW CONTINGENT DATA WORK SHAPES IDENTITY ATTRIBUTES IN COMPUTER VISION

In study two, I seek to understand how the identities of annotators influence the outcome of data annotation. In this study, the focus is on vendor contract workers. Contractors work for third party contracting companies that do outsourced labor for their clients—for the purposes of this study, technology companies. In particular, I am interested in contingent data work that involves annotating and/or collecting visual data for computer vision models to better understand the process of collection, annotation, the division of labor, the negotiations between contract workers and their clients, and the role of identity in annotation. Participants for this study will be both (1) employees of a contracting company, Envision Data (pseudonym), that provides data collection and annotation services, regardless of their specialization; and (2) freelance data workers who regularly provide data collection or annotation services on Upwork.

Findings from this study will also be analyzed alongside findings from Study One and Study Two to paint a clearer picture of how work at major tech companies and third-party annotation companies intersect. Like the prior two proposed studies, I will analyze interviews from participants from EnVision Data, including their

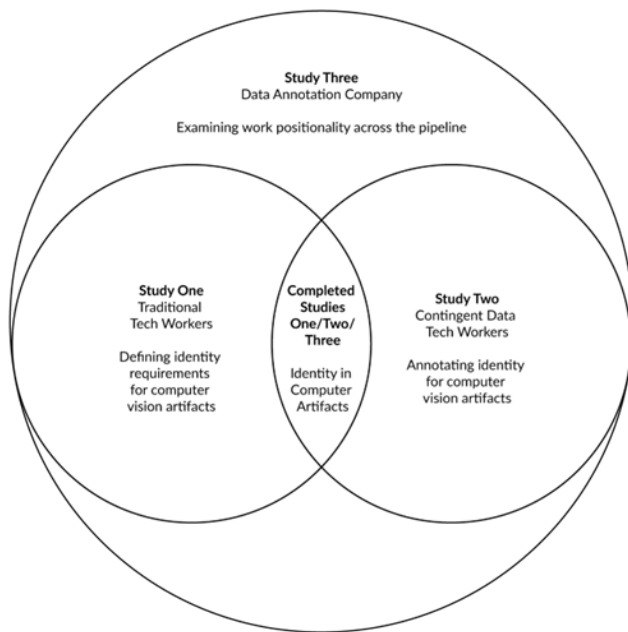


Figure 1: A figure showing how my completed work and my proposed studies will be integrated into my dissertation.

traditional tech workers, data annotators, and outsourced Upworkers. I will also analyze their documentation and my field notes, which include notes about observations and their artifacts. Results of this study will provide insight into how findings from Study One (on traditional tech workers) and Study Two (on data workers) look in practice, focusing on how positionality is negotiated and incorporated into the mission of the company and then enacted through the actual data work being done.

6 CONCLUSION

Human identity is central to human-centric computer vision, and decisions about how human identity is conceptualized in computer vision is driven by tech workers who hold their own positional values and perspectives. Through this dissertation, I argue that identity is conceptualized across three infrastructural layers of computer vision: (1) in the artifact layer, in the datasets and models identity is embedded in; (2) in the development layer, where classification ontology is defined and designed; and (3) in the annotation layer, where the ontology is applied. How identity is conceptualized in these three layers is reflective of an array of interconnecting positionalities and values.

Collectively, the six studies to be included in my dissertation will showcase how identity is embedded in each of the three layers and the role that worker positionality plays in shaping those conceptualizations. The outcome of this dissertation will be to provide computer vision developers and researchers with a more nuanced and complex perspective about identity—not only the implementation of identity categories in artifacts, but the role of positionality in shaping those categories and artifacts in the first place. My work

will assess computer vision infrastructures, not just as sociotechnical systems, but as inherited and interconnected layers with the goal of understanding how they reference, leverage, and constrain one another. I will connect the dots of positionality across the computer vision pipeline, from traditional tech worker to data worker to the final product, both dataset and model.

7 BIOGRAPHICAL SKETCH

I am a PhD candidate in the Information Science department at University of Colorado Boulder, advised by Jed Brubaker in the Identity Lab. My research focus is on how complex human identities, like gender and race, are classified and calcified for technological use, particularly in computer vision. My goal in understanding how and for what purposes human identity is embedded into technologies is to better improve representations in those technologies, and to consider whether identity should be used at all. My hope in attending the CSCW DC is to gain advice on improving my work to date, as well as career advice as I enter into my final year as a PhD student. More specifically, I seek feedback on data analysis. Given the open-ended and vast opportunities for data analysis and the presentation of my work, I seek advice on potentially fruitful directions and tensions that arise in how I approach presenting my work in my dissertation.

ACKNOWLEDGMENTS

This work was funded by a Microsoft Research PhD Fellowship.

REFERENCES

- [1] F. Anthias. 2008. Thinking through the lens of translocational positionality: an intersectionality frame for understanding identity and belonging. *Translocations: Migration and Social Change* 4, 1: 5–19. Retrieved April 4, 2022 from <https://repository.uel.ac.uk/item/8656x>
- [2] Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Benoit, Siham Tabik, Alberto Barbado, Salvador Garcia, Sergio Gil-Lopez, Daniel Molina, Richard Benjamins, Raja Chatila, and Francisco Herrera. 2020. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion* 58: 82–115. <https://doi.org/10.1016/j.inffus.2019.12.012>
- [3] Blaise Agüera y Arcas. 2017. Do algorithms reveal sexual orientation or just expose our stereotypes? Medium. Retrieved from <https://medium.com/@blaisea/do-algorithms-reveal-sexual-orientation-or-just-expose-our-stereotypes-d998f4df477>
- [4] Joy Buolamwini and Timnit Gebru. 2018. Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification *. Retrieved January 23, 2019 from <http://proceedings.mlr.press/v81/buolamwini18a/buolamwini18a.pdf>
- [5] Sixue Gong, Xiaoming Liu, and Anil K. Jain. 2019. DebFace: De-biasing Face Recognition. Retrieved December 3, 2019 from <http://arxiv.org/abs/1911.08080>
- [6] Foad Hamidi, Morgan Klaus Scheuerman, and Stacy M Branham. 2018. Gender Recognition or Gender Reductionism? The Social Implications of Automatic Gender Recognition Systems. In *2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*.
- [7] Os Keyes. 2018. The Misgendering Machines: Trans/HCI Implications of Automatic Gender Recognition. *Proceedings of the ACM on Human-Computer Interaction* 2, CSCW: 1–22. <https://doi.org/10.1145/3274357>
- [8] Brendan F. Klare, Mark J. Burge, Joshua C. Klontz, Richard W. Vorder Bruegge, and Anil K. Jain. 2012. Face recognition performance: Role of demographic information. *IEEE Transactions on Information Forensics and Security* 7, 6: 1789–1801. <https://doi.org/10.1109/TIFS.2012.2214212>
- [9] Sharan B. Merriam, Juanita Johnson-Bailey, Ming Yeh Lee, Youngwha Kee, Gabo Nteane, and Mazanah Muhamad. 2001. Power and positionality: Negotiating insider/outsider status within and across cultures. *International Journal of Lifelong Education* 20, 5: 405–416. <https://doi.org/10.1080/02601370120490>
- [10] Milagros Miceli and Julian Posada. 2021. Wisdom for the Crowd: Discursive Power in Annotation Instructions for Computer Vision. Retrieved September 14, 2021 from <https://arxiv.org/abs/2105.10990v1>

- [11.] Milagros Miceli, Martin Schuessler, and Tianling Yang. 2020. Between Subjectivity and Imposition: Power Dynamics in Data Annotation for Computer Vision. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2: 25. <https://doi.org/10.1145/3415186>
- [12.] Mei Ngan and Patrick Grother. 2015. Face Recognition Vendor Test (FRVT) - Performance of Automated Gender Classification Algorithms. <https://doi.org/10.6028/NIST.IR.8052>
- [13.] Kenny Peng, Arunesh Mathur, and Arvind Narayanan. 2021. Mitigating Dataset Harms Requires Stewardship: Lessons from 1000 Papers. Retrieved November 16, 2021 from <https://arxiv.org/abs/2108.02922v1>
- [14.] Inioluwa Deborah Raji, Timnit Gebru, Margaret Mitchell, Joy Buolamwini, Joonseok Lee, and Emily Denton. 2020. Saving Face: Investigating the ethical concerns of facial recognition auditing. In *AIES 2020 - Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 145–151. <https://doi.org/10.1145/3375627.3375820>
- [15.] Morgan Klaus Scheuerman, Alex Hanna, and Emily Denton. 2021. Do Datasets Have Politics? Disciplinary Values in Computer Vision Dataset Development. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2. <https://doi.org/10.1145/3476058>
- [16.] Morgan Klaus Scheuerman, Jacob M Paul, and Jed R Brubaker. 2019. How Computers See Gender: An Evaluation of Gender Classification in Commercial Facial Analysis and Image Labeling Services. 144: 33. <https://doi.org/10.1145/3359246>
- [17.] Morgan Klaus Scheuerman, Kandrea Wade, Caitlin Lustig, and Jed R. Brubaker. 2020. How We've Taught Algorithms to See Identity: Constructing Race and Gender in Image Databases for Facial Analysis. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW1.
- [18.] Sofia Marques da Silva and Joan Parker Webster. 2018. Positionality and Standpoint. *The Wiley Handbook of Ethnography of Education*, 501–512. <https://doi.org/10.1002/9781118933732.ch22>
- [19.] Steven Vallas and Juliet B. Schor. 2020. What do platforms do? Understanding the gig economy. In *Annual Review of Sociology*. *Annual Reviews*, 273–294. <https://doi.org/10.1146/annurev-soc-121919-054857>
- [20.] Tianlu Wang, Jieyu Zhao, Mark Yatskar, Kai-Wei Chang, and Vicente Ordonez. 2019. Balanced Datasets Are Not Enough: Estimating and Mitigating Gender Bias in Deep Image Representations. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*.